



GIRRAWEE HIGH SCHOOL
HALF YEARLY EXAMINATION

2006
MATHEMATICS
EXTENSION 1

*Time allowed - Two hours
(Plus 5 minutes' reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are on the laminated sheets supplied
- Board-approved calculators may be used.
- Each question attempted is to be returned on a *separate* piece of paper clearly marked Question 1 , Question 2 , etc. Each piece of paper must show your name.

Total marks – 98

Attempt Questions 1 – 5

All questions are NOT of equal value

Answer each question on a separate piece of paper clearly marked Question 1, Question 2 etc.

Each piece of paper must show your name.

Question 1 (17 Marks) Use a separate piece of paper **Marks**

- (a) Solve for x $\frac{x+2}{x-4} \leq 3$ **2**
- (b) Find the exact value of $\sin 75^\circ$ with a rational denominator **3**
- (c) Sketch the graph of $y = \sin 2x - 1$ $0 \leq x \leq 2\pi$ **3**
- (d) Find the acute angle between the lines $x - 3y + 5 = 0$ and $2x + y - 3 = 0$ **3**
- (e) Find the coordinates of the point that divides the interval
A(- 3, - 1) and B(6 , 10) internally in the ratio 2:1. **2**
- (f) Solve for x $3^x = 7$ to 2 decimal places **2**
- (g) Find $\lim_{x \rightarrow 0} \frac{\sin 5x}{3x}$ **2**

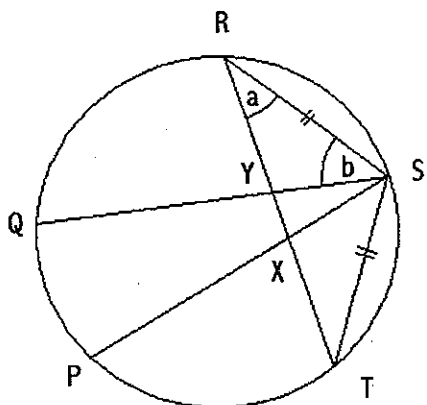
Question 2 (26 marks) Use a separate piece of paper

- (a) Differentiate the following
- (i) $y = x \ln x$ (ii) $y = \frac{\sin x}{e^{x^2}}$ **4**
- (iii) $y = \ln (\tan x)$ (iv) $y = \ln \frac{(x-1)}{(x-4)}$ **4**
- (v) $y = \cos^3 x$ **2**
- (b) Find
- (i) $\int \frac{x^2 + 2x}{x^3 + 3x^2 - 5} dx$ (ii) $\int x e^{2x^2} dx$ **4**
- (iii) $\int_0^{\frac{\pi}{12}} \sin^2 3x dx$ (iv) $\int_2^{10} \sqrt{2x+5} dx$ **6**
- (v) $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sec^2 \frac{x}{2} dx$ (vi) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x dx$ **6**

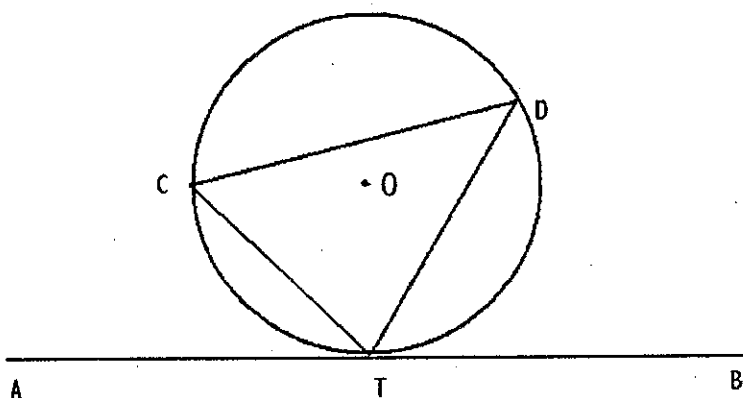
Question 3 (17 Marks) Use a separate piece of paper

Marks

- (a) Let PQRST be a circle such that $RS = ST$, QS meets RT at Y and PS meets RT at X, as in the diagram. Let $\angle YRS = a$ and $\angle QSR = b$



- | | | |
|-------|---|---|
| (i) | Copy the diagram and state why $\angle SYT = a + b$ | 2 |
| (ii) | Prove that $\angle RPQ = b$ | 2 |
| (iii) | Prove that $\angle RPS = a$ | 2 |
| (iv) | Prove that PQYX is a cyclic quadrilateral | 2 |
- (b) In the diagram below the tangent ATB meets the circle at T. CT is the chord from the point of contact. Prove the alternate segment theory. That is $\angle ATC = \angle CDT$



- | | | |
|-----|--------------------------------------|---|
| (c) | Prove by mathematical induction that | 4 |
|-----|--------------------------------------|---|

$$\frac{1}{2.3} + \frac{1}{3.4} + \frac{1}{4.5} + \dots + \frac{1}{(n+1)(n+2)} = \frac{n}{2(n+2)}$$

Question 4 (18 Marks) Use a separate piece of paper

Marks

- (a) Find the term independent of x in the following expression.
Leave your answer in unexpanded form.

3

$$\left(3x^3 - \frac{2}{x^2}\right)^{10}$$

- (b) (i) Give the expansion for $\tan (a + b)$

1

- (ii) Given that a , b and c are the three internal angles of a triangle, by using part (i) or otherwise prove that

$$\tan a + \tan b + \tan c = \tan a \tan b \tan c$$

3

- (c) The probability of passing a particular test is $\frac{3}{5}$. If 10 students sit the exam, what is the probability that

- (i) all 10 students pass

2

- (ii) at least 2 students pass

3

- (d) Using the letters of the name **JTVAJIRAJAH** find

- (i) The number of arrangements

2

- (ii) The number of arrangements with the **J**'s together

2

- (iii) The number of arrangements with the **R** and the **H** at the ends

2

Question 5 (20 Marks) Use a separate piece of paper

Marks

(a) Express $\cos x + \sin x$ in the form $R \cos (x - \alpha)$ clearly stating the values of both R and α .

3

(b) Hence sketch the graph of $y = \cos x + \sin x$ in the domain $0 \leq x \leq 2\pi$

4

(c) Use the t method, where $t = \tan \frac{\theta}{2}$, to solve the following. Give your answer in radians to 2 decimal places $0 \leq \theta \leq 2\pi$

4

$$4 \sin \theta + 3 \cos \theta = 2$$

(d) Find the volume when the curve $y = \cos x + \sec x$ is rotated about the x -axis between $x = 0$ and $\frac{\pi}{3}$

4

(e) (i) Differentiate the function $y = x \ln(x) - x$

2

(ii) Hence or otherwise find $\int_1^e \ln x dx$

3

EXTENSION 1 HALF YEARLY SOLUTIONS 2006.

Question 1 17 MARKS

(a) $\frac{x+2}{x-4} \leq 3 \quad x \neq 4$

$$(x-4)^2 \frac{x+2}{x-4} \leq 3(x-4)^2$$

$$(x-4)(x+2) \leq 3(x-4)^2$$

$$0 \leq (x-4)\{3(x-4) - (x+2)\}$$

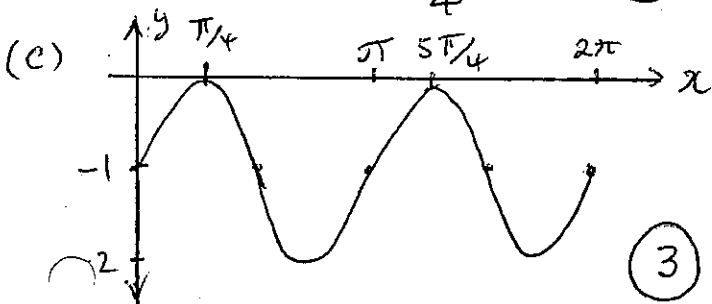
$$0 \leq (x-4)(2x-14)$$

$$\therefore x < 4 \quad x \geq 7 \quad (2)$$

(b) $\sin 75^\circ = \sin(45+30)$
 $= \sin 45 \cos 30 + \cos 45 \sin 30$
 $= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$

$$= \frac{\sqrt{3}+1}{2\sqrt{2}}$$

$$= \frac{\sqrt{6}+\sqrt{2}}{4} \quad (3)$$



(d) $x-3y+5=0 \Rightarrow 3y=x+5$
 $\therefore m_1 = 1/3 \quad y = 1/3x + 5/3$

$$2x+y-3=0 \Rightarrow y = -2x+3$$

$$\therefore m_2 = -2$$

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$= \frac{1/3 + 2}{1 - 2/3}$$

$$= 7$$

$$\theta = 81.87^\circ, 81^\circ 52' 11.63'' \quad (3)$$

(e) $\frac{6 \times 2 + -3 \times 1}{2+1}, \frac{10 \times 2 + -1 \times 1}{2+1}$

$$(3, \frac{19}{3}) \quad (2)$$

(f) $3^x = 7$

$$\ln 3^x = \ln 7$$

$$x \ln 3 = \ln 7$$

$$x = \frac{\ln 7}{\ln 3}$$

$$x = 1.77 \quad (2 \text{ D.P.}) \quad (2)$$

(g) $\lim_{x \rightarrow 0} \frac{\sin 5x}{3x} = \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \cdot \frac{5}{3}$

$$= 1 \cdot \frac{5}{3}$$

$$= \frac{5}{3} \quad (2)$$

Question 2. (26 MARKS)

(a) (i) $y = x \ln x$

$$\frac{dy}{dx} = x \cdot \frac{1}{x} + 1 \cdot \ln x$$

$$= 1 + \ln x \quad (2)$$

(ii) $y = \frac{\sin x}{e^{x^2}}$

$$\frac{dy}{dx} = v \cdot \frac{du}{dx} - u \cdot \frac{dv}{dx}$$

$$= \frac{e^{x^2} \cdot \cos x - \sin x \cdot (2x e^{x^2})}{(e^{x^2})^2}$$

$$= \frac{\cos x - 2x \sin x}{e^{x^2}} \quad (2)$$

(iii) $y = \ln(\tan x)$

$$\frac{dy}{dx} = \frac{\sec^2 x}{\tan x} \quad (2)$$

$$14) y = \ln \frac{x-1}{x-4}$$

$$y = \ln(x-1) - \ln(x-4)$$

$$\frac{dy}{dx} = \frac{1}{x-1} - \frac{1}{x-4} \quad (2)$$

$$15) y = \cos^3 x$$

$$\frac{dy}{dx} = \frac{d\cos^3 x}{d\cos x} \cdot \frac{d\cos x}{dx}$$

$$= 3\cos^2 x \cdot -\sin x$$

$$= -3\cos^2 x \sin x \quad (2)$$

$$16) (i) I = \int \frac{x^2 + 2x}{x^3 + 3x^2 - 5} dx$$

$$= \frac{1}{3} \int \frac{3x^2 + 6x}{x^3 + 3x^2 - 5} dx$$

$$= \frac{1}{3} \ln |x^3 + 3x^2 - 5| + C \quad (3)$$

$$(ii) I = \int x e^{2x^2} dx$$

$$= \frac{1}{4} \int 4x e^{2x^2} dx$$

$$= \frac{1}{4} e^{2x^2} + C \quad (3)$$

$$(iii) I = \int_0^{\pi/12} \sin^2 3x dx$$

$$= \frac{1}{2} \int_0^{\pi/12} 1 - \cos 6x dx$$

$$= \frac{1}{2} \left[x - \frac{\sin 6x}{6} \right]_0^{\pi/12}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{12} - \frac{1}{6} \right) - (0) \right]$$

$$= \frac{1}{24} (\pi - 2) \quad (3)$$

$$(iv) I = \int_2^{10} (2x+5)^{1/2} dx$$

$$= \left[\frac{2}{3} \cdot \frac{1}{2} (2x+5)^{3/2} \right]_2^{10}$$

$$= \frac{1}{3} [(125) - (27)]$$

$$= \frac{98}{3} \quad (3)$$

$$(v) I = \int_{\pi/3}^{\pi/2} \sec^2 \frac{x}{2} dx$$

$$= \left[2 \tan \frac{x}{2} \right]_{\pi/3}^{\pi/2}$$

$$= 2 \left\{ \left(\tan \frac{\pi}{4} \right) - \left(\tan \frac{\pi}{6} \right) \right\}$$

$$= 2 \left(1 - \frac{1}{\sqrt{3}} \right) \quad (3)$$

$$(vi) I = \int_{-\pi/2}^{\pi/2} \sin x dx \quad (3)$$

$$= 0 \quad (\sin x \text{ is an odd fn.})$$

Question 3. (17 MARKS)

(a) (i) 1 mark for diagram

$\angle SYT$ IS THE EXTERNAL ANGLE TO $\triangle SYR$.

$$\therefore \angle SYT = \angle SRY + \angle RSY \quad (2)$$

$$= a + b$$

EXTERNAL $\angle$ = TO THE SUM OF THE INTERIOR OPPOSITE $\angle$ 'S.

$$(ii) \angle RSQ = \angle RPQ$$

ANGLE STANDING ON THE SAME CHORD OR ARC (RQ) ARE EQUAL IN THE SAME SEGMENT

$$\therefore \angle RPQ = b \quad (2)$$

(iii) $\triangle RST$ IS ISOSCELES

$$RS = ST$$

$$\therefore \angle SRT = \angle STR \quad (2)$$

$$\therefore \angle STR = a$$

$$\therefore \angle SPR = a$$

ANGLES STANDING ON THE SAME ARC (RS) IN THE SAME SEGMENT ARE EQUAL. (2)

$$(iv) \angle RPQ = b$$

$$\angle RPS = a$$

$$\therefore \angle QPS = a + b$$

$$(\angle RPQ + \angle RPS)$$

$$\angle SYT = a + b \text{ (PART (i))}$$

$$\therefore \angle QPS = \angle SYT$$

THAT IS EXTERIOR $\angle SYT$ OF QUADRILATERAL PQYX (2)

IS EQUAL TO THE INTERIOR OPPOSITE ANGLE $\angle QPS$

$\therefore$ PQYX IS A CYCLIC QUAD.

b) CONSTRUCT THE DIAMETRE FROM T THROUGH O MEETING THE CIRCLE AT F. JOIN C TO F

$$\angle ATC + \angle CTF = 90^\circ$$

(TANGENT $\perp$ TO RADIUS)

$$\angle CTF + \angle CFT = 90^\circ$$

($\angle TCF = 90^\circ$, ANGLE IN SEMI-CIRCLE)

$$\therefore \angle ATC = \angle CFT$$

$$\text{But } \angle CFT = \angle CDT$$

(ANGLES STANDING ON THE SAME ARC IN THE SAME SEGMENT ARE EQUAL)

$$\therefore \angle ATC = \angle CDT \quad (5)$$

$$(c) \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} \dots \frac{1}{(n+1)(n+2)}$$

$$= \frac{n}{2(n+2)}$$

Step 1 Prove true for $n=1$

$$LHS = \frac{1}{2 \times 3} = \frac{1}{6}$$

$$RHS = \frac{1}{2(1+3)} = \frac{1}{6}$$

$$LHS = RHS \text{ TRUE FOR } n=1$$

Step 2 Assume true for $n=k$ (1)

$$\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{(k+1)(k+2)} = \frac{k}{2(k+2)}$$

Step 3 Prove true for $n=k+1$

$$RHS = \frac{k+1}{2(k+3)}$$

$$LHS = \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{(k+1)(k+2)} + \frac{1}{(k+2)(k+3)}$$

$$= \frac{k}{2(k+2)} + \frac{1}{(k+2)(k+3)}$$

$$= \frac{k(k+3) + 2}{2(k+2)(k+3)}$$

$$= \frac{k^2 + 3k + 2}{2(k+2)(k+3)}$$

$$= \frac{(k+2)(k+1)}{2(k+2)(k+3)}$$

$$= \frac{k+1}{2(k+3)}$$

$$= RHS$$

$\therefore$ TRUE FOR $n=k+1$

Step 4. Proved true for $n=1$

Step 3 implies it true for $n=1$
true for $n=2, 3, 4, 5 \dots$

$\therefore$ BY THE PRINCIPLE OF M.I
TRUE FOR ALL n . (1)

Question 4 (18 MARKS)

$$1) \left(3x^3 - \frac{2}{x^2}\right)^{10} = {}^{10}C_0 (3x^3)^{10} + {}^{10}C_1 (3x^3)^9 \left(\frac{-2}{x^2}\right)$$

$$\dots + {}^{10}C_n (3x^3)^{10-n} \left(\frac{-2}{x^2}\right)^n$$

$${}^{10}C_n (3x^3)^{10-n} \left(\frac{-2}{x^2}\right)^n = {}^{10}C_n 3^{10-n} (-2)^n x^{30-3n-2n}$$

INDEPENDENT OF x

$$x^{30-5n} = x^0$$

$$30-5n = 0$$

$$n = 6$$

$$\text{TERM } {}^{10}C_7 3^4 2^6 \quad (3)$$

$$b) (i) \tan(a+b) = \frac{\tan a + \tan b}{1 - \tan a \tan b}$$

(1)

$$(ii) a+b+c = \pi$$

$$a+b = \pi - c$$

$$\tan(a+b) = \tan(\pi - c)$$

$$\frac{\tan a + \tan b}{1 - \tan a \tan b} = \frac{\tan \pi - \tan c}{1 + \tan \pi \tan c}$$

$$\frac{\tan a + \tan b}{1 - \tan a \tan b} = \frac{-\tan c}{1} \quad (3)$$

$$\tan a + \tan b = -\tan c (1 - \tan a \tan b)$$

$$= -\tan c + \tan a \tan b \tan c$$

$$\tan a + \tan b + \tan c = \tan a \tan b \tan c$$

$$c) (i) \left(\frac{3}{5} + \frac{2}{5}\right)^{10} = {}^{10}C_0 \left(\frac{3}{5}\right)^{10} \quad (2)$$

$$= 0.006 \text{ (3 D.P.)}$$

$$(ii) P(\text{at least 2}) = 1 - \{P(1 \text{ pen}) + P(0 \text{ pen})\}$$

$$= 1 - \left({}^{10}C_9 \left(\frac{3}{5}\right)^1 \left(\frac{2}{5}\right)^9 + {}^{10}C_{10} \left(\frac{3}{5}\right)^0 \left(\frac{2}{5}\right)^{10} \right)$$

$$= 1 - (1.57 \times 10^{-3} + 1.05 \times 10^{-4})$$

$$= 0.998 \quad (3 \text{ D.P.})$$

(3)

$$(d) (i) \# \text{ arrangements} = \frac{11!}{3! 3! 2!}$$

$$(3 \text{ J's}, 3 \text{ A's}, 2 \text{ I's}) \quad (2)$$

$$= 554400$$

$$(ii) \# \text{ arrangement} = \frac{9!}{3! 2!} \quad (1)$$

$$\text{REGARD J's AS ONE LETTER.} \quad (2)$$

$$= 5040$$

$$(iii) \# \text{ arrangement} = \frac{2 \times 9!}{3! 3! 2!} \quad (2)$$

$$= 1680$$

QUESTION 5 (22 MARKS)

$$(a) R \cos(x+\alpha) = R \cos x \cos \alpha + R \sin x \sin \alpha$$

$$\cos x = R \cos \alpha \cos x \quad (A) \quad (1)$$

$$1 = R \cos \alpha \quad (A)$$

$$\sin x = R \sin \alpha \sin x \quad (B)$$

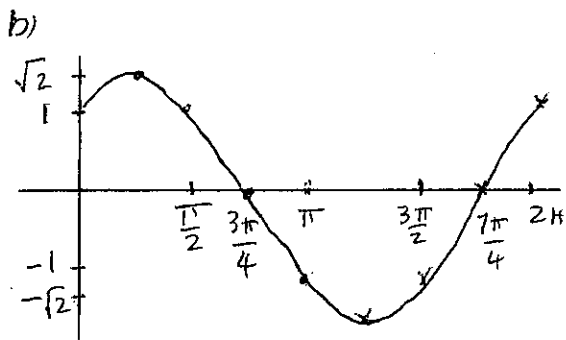
$$1 = R \sin \alpha \quad (B)$$

$$\frac{B}{A} \quad \frac{R \sin \alpha}{R \cos \alpha} = 1 \quad \tan \alpha = 1 \quad (1) \quad \alpha = 45^\circ$$

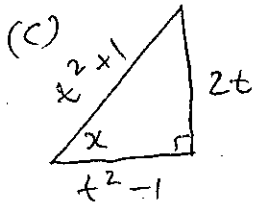
$$B^2 + A^2 = R^2 \sin^2 \alpha + R^2 \cos^2 \alpha = 2$$

$$R^2 (\sin^2 \alpha + \cos^2 \alpha) = 2$$

$$R = \sqrt{2} \quad (1)$$



$$y = \cos x + \sin x.$$



$$\cos x = \frac{t^2-1}{t^2+1}$$

$$\sin x = \frac{2t}{t^2+1}$$

$$4 \sin \theta + 3 \cos \theta = 2$$

$$4 \frac{2t}{t^2+1} + 3 \frac{t^2-1}{t^2+1} = 2$$

$$8t + 3t^2 - 3 = 2t^2 + 2$$

$$t^2 + 8t - 5 = 0$$

$$t = \frac{-8 \pm \sqrt{84}}{2}$$

$$\tan \frac{x}{2} = 0.5826, -8.5826$$

$$\frac{x}{2} = 0.5275, 3.6691, 1.6870, 4.8284$$

$$x = 1.055^\circ, 3.374^\circ$$

(4)

(d) $y = \cos x + \sec x$

$$V = \pi \int_0^{\pi/3} y^2 dx$$

$$= \pi \int_0^{\pi/3} \cos^2 x + \sec^2 x + 2 dx$$

$$= \pi \int_0^{\pi/3} \frac{1}{2} \cos 2x + \sec^2 x + \frac{5}{2} dx$$

$$= \pi \left[\frac{\sin 2x}{4} + \tan x + \frac{5x}{2} \right]_0^{\pi/3}$$

$$= \pi \left[\left(\frac{\sqrt{3}}{8} - \sqrt{3} + \frac{5\pi}{6} \right) - (0) \right]$$

$$= \pi \left[\frac{5\pi}{6} - \frac{7\sqrt{3}}{8} \right] \mu^3 \quad (4)$$

(e) (i)

$$y = x \ln x - x$$

$$\frac{dy}{dx} = x \cdot \frac{1}{x} + 1 \ln x - 1$$

$$= \ln x$$

(2)

(ii) $I = \int_1^e \ln x dx$

$$= \left[x \ln x - x \right]_1^e$$

$$= \left[(e - e) - (-1) \right]$$

$$= 1 \mu^2$$

(3)

